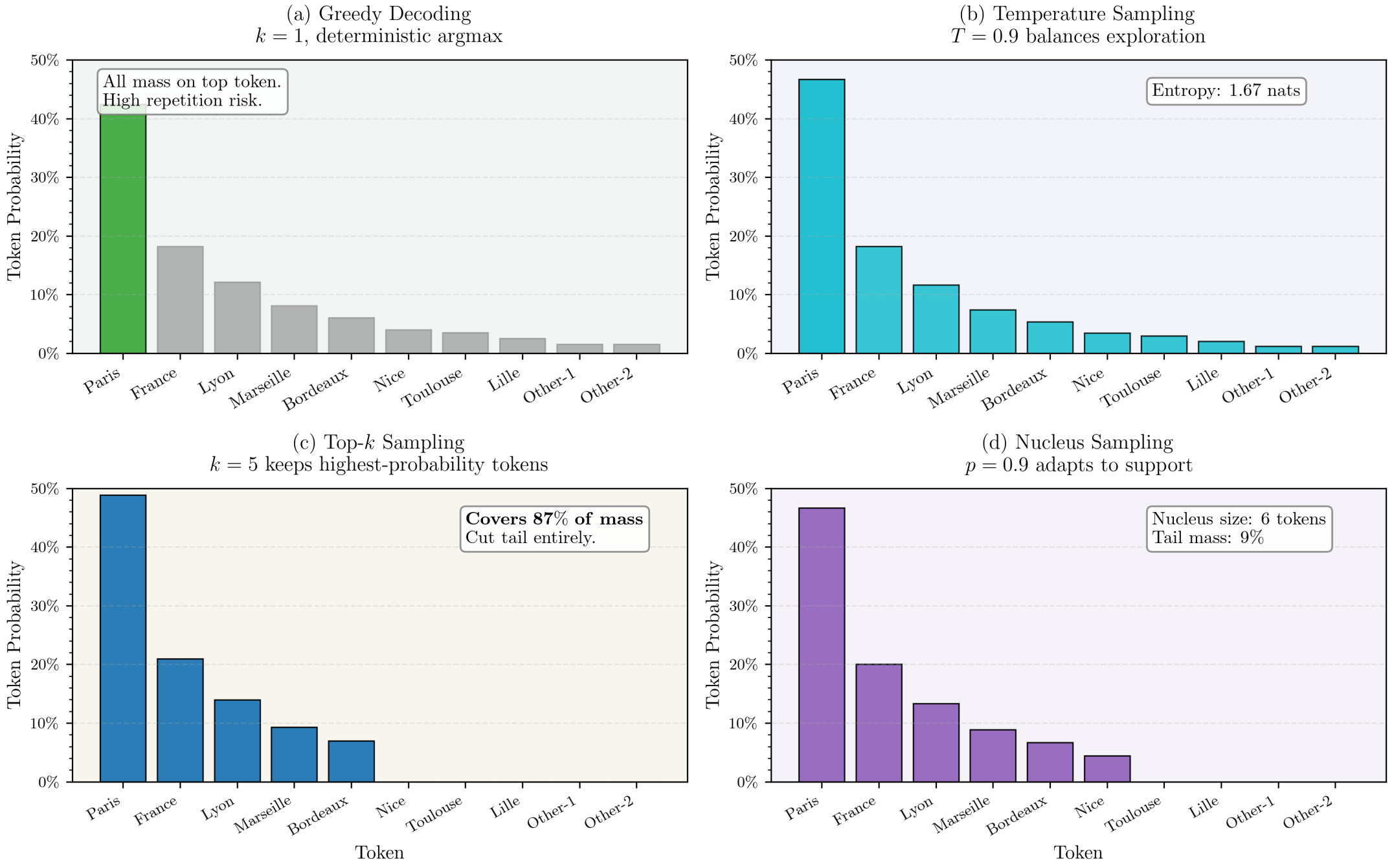
(a) Greedy Decoding
$k = 1$, deterministic argmax
All mass on top token.
High repetition risk.
(b) Temperature Sampling
$T = 0.9$ balances exploration
Entropy: 1.67 nats
(c) Top-k Sampling
$k = 5$ keeps highest-probability tokens
Covers 87% of mass
Cut tail entirely.
(d) Nucleus Sampling
$p = 0.9$ adapts to support
Nucleus size: 6 tokens
Tail mass: 9%
Token Probability
0%
10%
20%
30%
40%
50%
Paris
France
Lyon
Marseille
Bordeaux
Nice
Toulouse
Lille
Other-1
Other-2
Token